CITIZEN SCIENCE SUPPLEMENT for Flat Tree Oyster (*Isognomon alatus*) search

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Helpful apps to download and how to use them:

Google Earth Pro (for computer) – Use this app to explore aerial photo maps of your area and pre-pick sites to sample. Make sure the app is set to display latitude and longitude in "decimal degrees." You can drop a "pin" to mark where you plan to sample.

Google Earth (for phone) – When you open this app outdoors and click the crosshairs icon it will use your phone's GPS to find and display the latitude and longitude of where you're standing. Write these numbers on your oyster survey datasheet where it says "Lat, Long:". The number followed by an N is your **latitude** (how many degrees north of the equator you are) and the number followed by a W is your **longitude** (how many degrees west of the prime meridian you are).

iNaturalist (website / app) – Create an account so you can upload photos of species you observe in the world. Real people and AI will help you identify those species, and your observations will contribute to a scientific record of what species are found where in the world. iNaturalist also functions as a social network; you can see what your friends post. My inaturalist handle is "jdouglass" if you want to follow my postings.

Seek (phone app by iNaturalist) – This simple app lets you hover your phone camera over a species and uses AI trained on iNaturalist images to deduce what species you're looking at. Make sure you set it to display scientific names because common names can be ambiguous. Seek is quick and easy but it doesn't create a permanent record that can be reviewed by real people later, so it's not as useful for serious science.

Tide and Weather apps – There are tons of apps that can give coastal weather and tide forecasts. If you want to survey the "intertidal zone" without having to stick your head underwater, you should plan to survey when the tide level is low.

Science Primer:

Scientific Names – To keep track of the millions of species of life on earth and how they're all related, scientists use a hierarchical classification system based on the evolutionary tree. From broadest to narrowest, the categories are Domain, Kingdom, Phylum, Class, Order, Family, Genus, and Species. When we write the "scientific name" of a species, like *Homo sapiens*, we write the name of both the genus (Homo) and the species within that genus (sapiens). On the oyster datasheet, some of the common bivalves are written at the species-specific level, while other things are listed as broader categories.

Common Names – While each species on earth has just one scientific name, it may be known by multiple, unofficial "common names." Also, the same common names may be used for several different species, leading to confusion.

Intertidal and Subtidal – The intertidal zone is the part of the seafloor that is exposed to the air at low tides and covered by water at high tides. The subtidal zone is the part of the seafloor that is deep enough to still be covered by water even during the lowest tides. Only species hardy enough to tolerate periodic exposure to air can inhabit the intertidal zone, so it is usually less diverse than the subtidal zone. That's why surveyors are asked to specify whether they searched both the intertidal and subtidal zones, or just the intertidal zone.

Hard Substrate- Substrate refers to the seabed and the structures on it to which life might attach. This oyster survey targets "hard" substrates which are things like rock, concrete, shell piles, docks/pilings, and mangrove roots; all the things that oysters could attach to. The survey ignores sand and mud substrates. So when you rate the coverage of oysters and other epibiota at your site, you are just

Sessile Epibiota / Epibionts / Epifauna – Sessile means life that stays in one place and doesn't move. All plants are sessile, and in the ocean many animals, such as oysters, corals, and barnacles, are also sessile. Epibiota / Epibionts means life forms that live attached to the bottom. Epifauna refers specifically to animals that live on the bottom. E.g., oysters and seaweeds would both count as epibiota, but only oysters would count as epifauna because seaweeds aren't animals.

Braun-Blanquet surveys:

The idea of a Braun-Blanquet (BB) survey is to identify what species are present in an area and approximately how abundant each species is. It's based on a visual estimate of how much space the species covers. Usually the estimate is made within a closely defined area like a quadrat frame, but in this case you're just going to do it for the entire site (or rather, for all the hard substrates within the site). The table below gives some visual examples to complement the BB scores 0.1 – 5. If you're having trouble picking between two BB scores, go for the lower one to be conservative. In the picture of a seawall below, *Crassostrea virginica*, *Isognomon alatus*, and Chlorophyta would each get a BB score of 2, and Rhodophyta would get a BB score of 1. There are some limpets and other snails in the picture but they don't get recorded because this survey is only for sessile (not mobile) organisms.



Table. 1 Visual key to Braun-Blanquet categories.

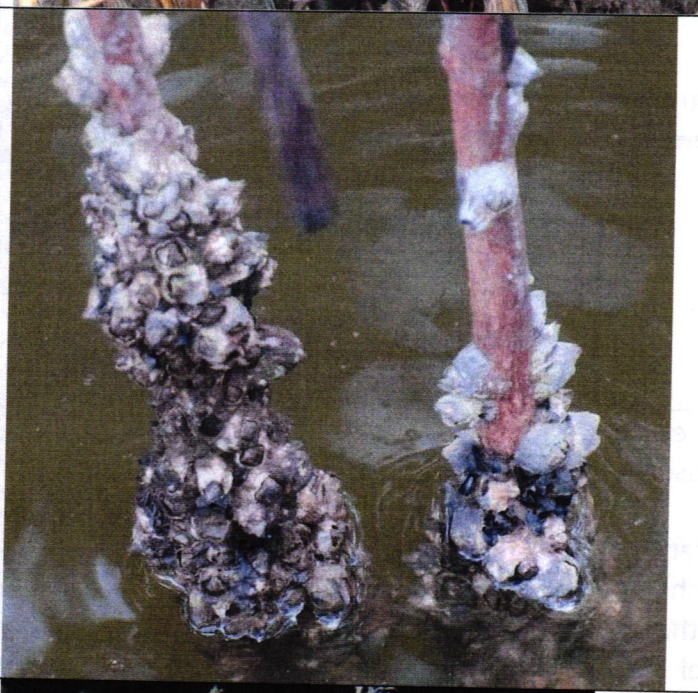
BB Score	Description	Graphic
0.1	Very rare; only a few individuals encountered in a large search area	
0.5	Rare; sparsely present but with much less than 5% cover	
1	Common but covering less than 5% of the relevant substrate	
2	Any number of individuals covering 5-25% of the relevant substrate	
3	Any number of individuals covering 25-50% of the relevant substrate	
4	Any number of individuals covering 50-75% of the relevant substrate	
5	Any number of individuals covering >75% of the relevant substrate	

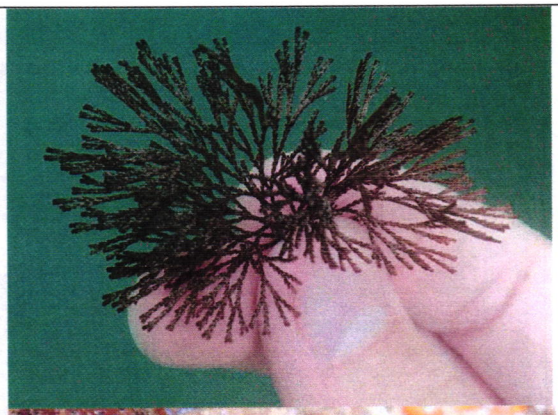
Species Identification- Below is a table of images to help with identification of some of the common sessile epibiota of intertidal and shallow subtidal hard substrate habitats in Florida. Some key things to look for with respect to the bivalves are:

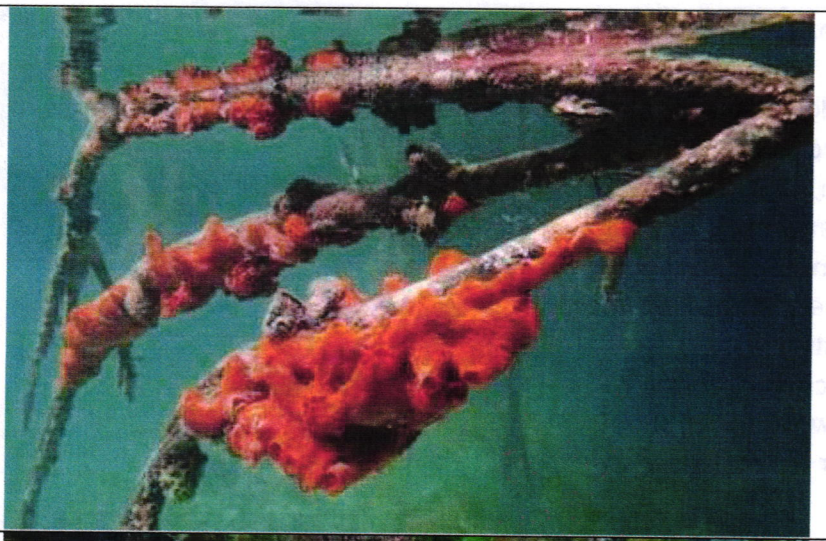
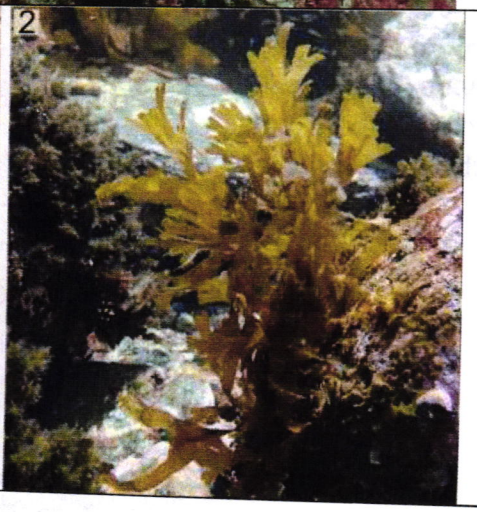
1. How are they attached? *Crassostrea* oysters are **cemented** to the substrate or each other; the left half of their shell is literally fused to whatever they're growing on. In contrast, *Isognomon* oysters are attached more like mussels, with "**byssal threads**." You can wiggle them like a loose tooth.
2. How are they shaped and textured? In general, oysters are rough and irregularly shaped, while mussels are smooth or finely grooved and have a consistent teardrop shape. Tree oysters are thin and wafer-like, neither as rough as regular oysters nor as smooth as mussels.

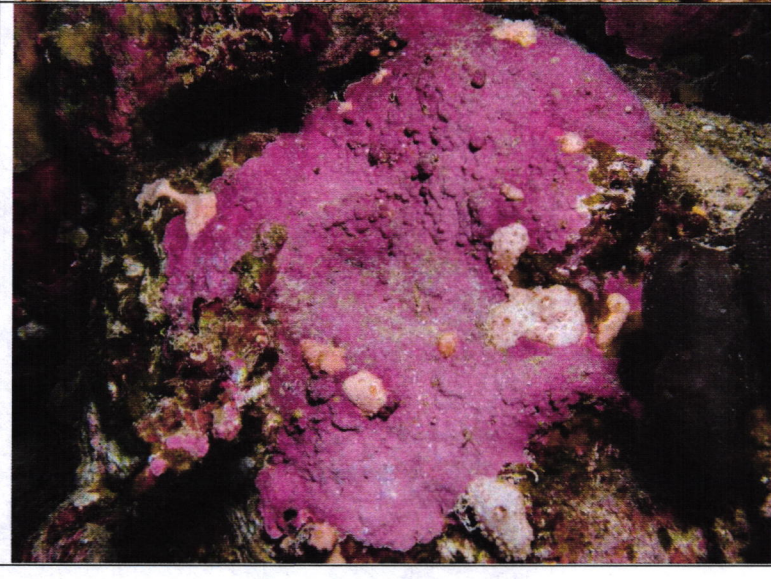
Crassotrea virginica (Eastern Oyster) *Cemented on *Rough, irregular shell shapes	
Isognomon alatus (Flat Tree Oyster) *Byssal attachment	

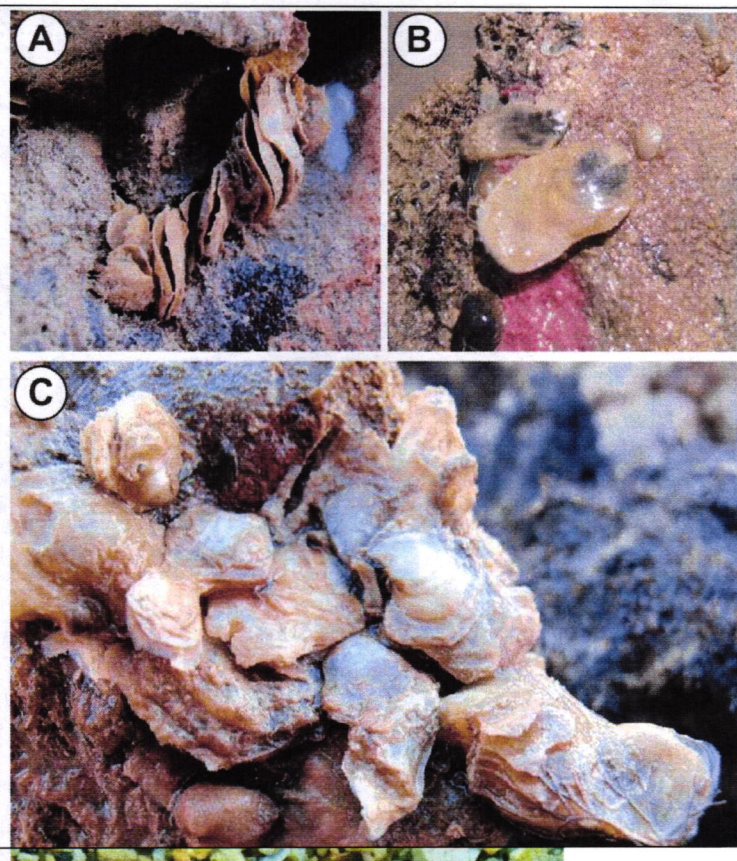
Perna viridis (Green Mussel) *Smooth and vivid jade color *Byssal attachment	
Ischadium recurvum (Hooked Mussel) *Very curved *Pronounced grooves *Byssal attachment	
Brachiodontes exustus (Scorched Mussel) *Smaller, lighter colored (often yellowish), and finer grooves than hooked mussel *Byssal attachment	

Geukensia granosissima (Southern Ribbed Mussel) *Shoe-shaped, often half-buried in mud of saltmarshes *Byssal attachment	
Cirripedia (Barnacles) *Cemented *Teepee shaped exoskeleton with "doors" at top	
Leptothecata (Lacy Hydroids) *Usually pale yellow or brown, wirey branching structure. Has tiny polyps (distant relative of corals and jellyfish)	

Bryozoa (Bryozoans) *Colonies made of tiny, fused compartments with a little openings. *Branching forms can be mistaken for red algae, but they are more crusty. *Encrusting forms grows as rough sheets over the substrate.	 
Ascidea (Sea Squirts / Tunicates) *May be grape-like individuals or colonies forming rubbery mats. *Rubbery; not spongy *	

Porifera (Sponges) *Spongy texture *Many colors	
Rhodophyta (Red Macroalgae) *Rarely bright red- usually more like dirty maroon tumbleweeds	
Phaeophyta (Brown Macroalgae) *Golden brown fronds or filaments	 

Chlorophyta (Green Macroalgae) *Green- Many shapes and textures but often mossy, filamentous, or like thin leaves of lettuce	
Corallinales (Coralline red algae) *Technically it's red macroalgae, but it looks like hard crust of pink or purple paint	

Isognomon bicolor (Two-toned tree oyster) *Rough little shells, more irregular than *I. alatus*	
Isognomon radiatus (Radial Purse Oyster) *Light shell color with distinct radial color banding *Often elongated shape	

Florida *Isognomon*

Text by Marlo F. Krisberg Aug 2012

Isognomon alatus
(Gmelin, 1791)



Obliquely fan-shaped.

Exterior smoothish but often with irregular growth lines and flaky concentric lamellations near the margins.

Exterior color drab purplish-gray to dirty gray.

Isognomon bicolor
(C. B. Adams, 1845)



Irregularly oval .

Exterior is coarsely lamellate with overlapping ridges that may be elevated, thin and usually chipped along the edges.

Exterior color ranges from dirty gray to yellowish-brown or purplish-brown.

Isognomon radiatus
(Anton, 1838)



Obliquely oval often arched to the left or right .

Exterior smoothish, but often with low, indistinct ribs.

Exterior color consistent, tan to yellowish-white with several widening pale purple or reddish-brown radial rays.